Smart Vacuum Cleaner

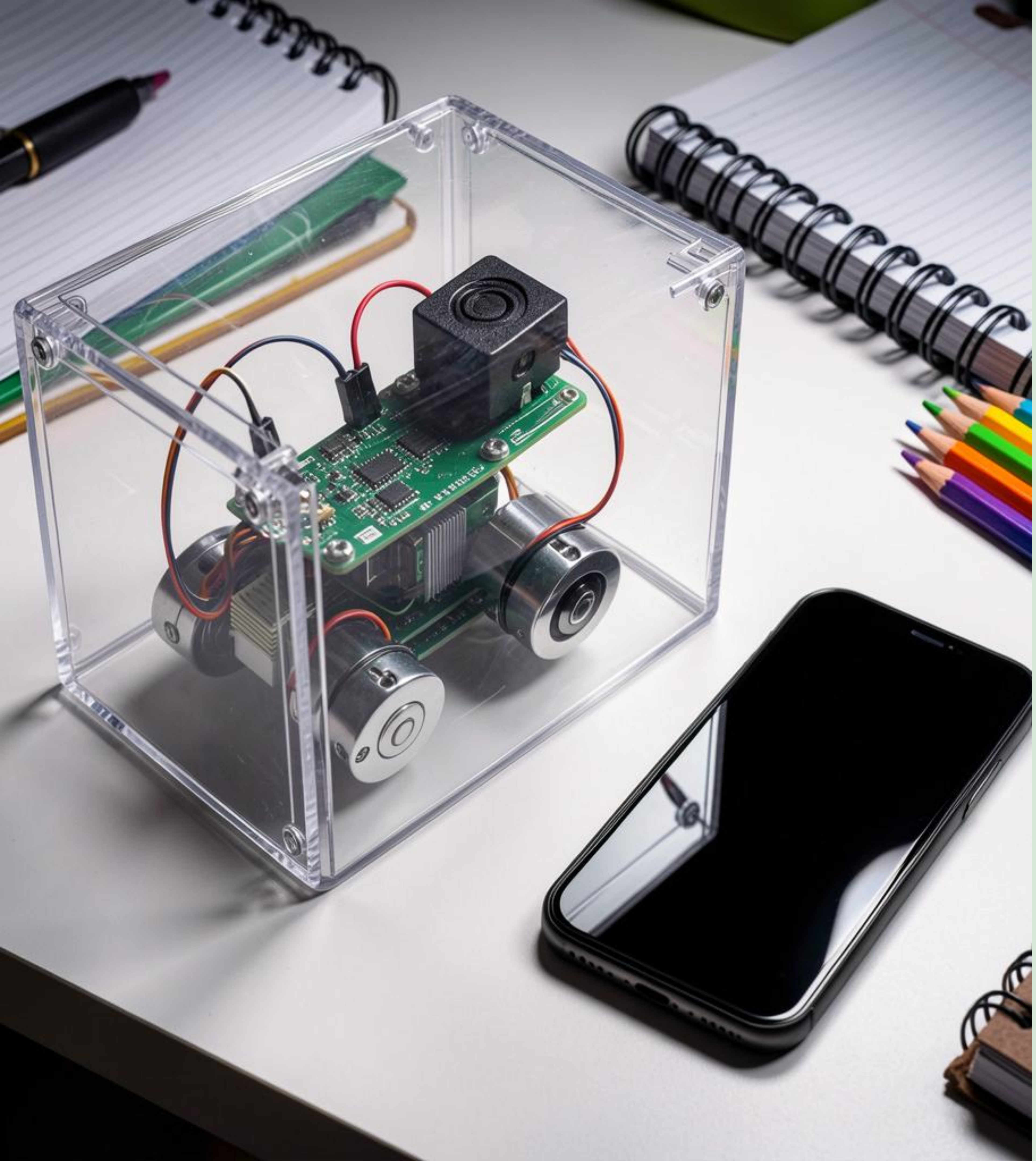
IoT robot using Arduino Uno for VIDYA Tech fest 2026-27

Introduction

Cleaning matters for health and comfort — classrooms and homes benefit from regular upkeep.

Smart solutions save time and effort by automating routine cleaning, making spaces healthier and more organized.

This project introduces an *Arduino-based robotic cleaner* that combines obstacle sensing and smartphone control for easy, student-friendly demonstrations.



01

Project Overview

Purpose & Goals

Build a compact, mobile vacuum robot controllable via smartphone that also avoids obstacles automatically.

Key aims: **reduce manual effort**, save time, teach IoT basics, and create a practical exhibition demo using real components like *Arduino Uno* and ultrasonic sensing.

Problem Statement

Manual cleaning is time-consuming and misses hard-to-reach spots under furniture.

Dust affects health and requires repetitive effort; students need an affordable, educational robot that shows how *automation* improves daily tasks.

Key Features

Smart vacuum moves in four directions via smartphone and cleans with a powerful fan.

Automatic obstacle avoidance using *ultrasonic sensing* keeps it safe and efficient.

Compact, battery-powered design makes it portable and ideal for school demos. **Wireless control** and **real-time response** are core highlights.

02

Hardware & Software

Components List

Nord MCU, HC-SR04 ultrasonic sensor, HC-05 Bluetooth module, L298N motor driver, DC motors, vacuum fan, rechargeable battery, wheels and chassis.

Each part is chosen for ease of use, availability, and clear educational value.

Circuit & Flow

Smartphone → Bluetooth (HC-05) → *Arduino Uno* → L298N → DC motors & vacuum fan.

Ultrasonic sensor feeds distance data to Arduino for obstacle detection and autonomous maneuvers. **Simple block wiring** keeps the circuit student-friendly.

Mobile Control

Mobile app provides directional buttons: ↑ ↓ ← → and Stop.

Commands sent via Bluetooth control motors; Arduino translates commands into motor driver signals while handling sensor-based avoidance. *Intuitive interface* for live demos.

Conclusions

The project demonstrates practical IoT, robotics, and embedded systems in a student-friendly build.

It saves time, reduces effort, and teaches core concepts like wireless control and sensor feedback.

Perfect for a hands-on science exhibition showing automation made simple.

Thanks!